Problem set

Exercise 1. Consider the following copula models:

i) Clayton copula with $\theta = 10$.

ii) Survival Clayton copula with $\theta = 5$

iii) Gaussian copula with $\rho = -0.7$

iv) Student's t copula with $\rho = -0.7$ and $v = 5$

v) FGM copula with parameter $\lambda = -0.5$

vi) FGM copula with parameter $\lambda = 0$

Answer the following questions:

1.a) Obtain 100,000 simulated bivariate dependent U(0,1) draws for each copula model. Make a plot of the dependent U (0,1) points from each copula (a total of six graphics). Obtain Pearson's correlation, Spearman's correlation and Kendall's tau for each case.

1.b) Assume the return of stock 1 is N (0,1) distributed and the return of stock 2 is Student t (with v=6) distributed. Given each copula in the previous section, compute the mean, standard deviation, skewness and kurtosis for the return portfolio containing 40% of stock 1 and 60% of stock 2. Obtain also the VaR (quantiles) at levels of 1% and 5%. Finally, obtain the correlation matrix corresponding to the six previous portfolio returns (i.e., a different portfolio for a different copula model). Plot the histogram of each of them joint with Normal PDF with the same mean and standard deviation of the portfolio return for each case.

1.c) Assume your stock return observations just the simulated returns of each stock in the previous section under the Gaussian copula with $\rho = -0.7$. Try to estimate a Gaussian copula by using the IMF method. Estimate the marginals by using the EDF method.

1.d) Assume a copula mixture with a probability of 0.6 for the Clayton copula with $\theta = 10$ and 0.4 for the survival Clayton with $\theta = 5$. Obtain 100,000 simulated bivariate dependent U (0,1) draws for this copula model. Assume again the N (0,1) marginal distribution for each stock return. Try to estimate a t copula by using the IMF method. Estimate the marginals by using the EDF method. Hint: see for the simulation method the Excel file: "**MixClayClay180.xlsx**".

Exercise 2. Consider the three daily return series from the Excel spreadsheet "**crypto.xlsx**".

2.1) Obtain the descriptive statistics for all series (mean, standard deviation, skewness and kurtosis), and also the sample correlation matrix.

2.2) Try to estimate the parameters of the DCC model by using the trivariate Gaussian-copula under alternative estimates for the marginal distributions:

a) The EDF as the estimation of the marginal distributions the model in the previous section. Make a plot of the three dynamic correlation series.

b) Estimate a GARCH-N model to each series by assuming a zero mean for the returns.

c) Estimate a GJR-N model to each series by assuming a zero mean for the returns.

2.3) Compare the three dynamic correlation series obtained in each case in section 2.2).

2.4) Estimate the parameters of the DCC model under the trivariate Student's t copula using EDF estimates for the marginal distributions.

Exercise 3. The CDF of the conditional copula for a bivariate Normal mixture (NM) copula is easily obtained from the bivariate Gaussian copula, and it is given by

$$C^{NM}\left(u_1 \middle| u_2\right) = \pi \times \Phi\left(\frac{\Phi^{-1}\left(u_1\right) - \rho_1\, \Phi^{-1}\left(u_2\right)}{\sqrt{1-\rho_1^2}}\right) + \left(1-\pi\right) \times \Phi\left(\frac{\Phi^{-1}\left(u_1\right) - \rho_2\, \Phi^{-1}\left(u_2\right)}{\sqrt{1-\rho_2^2}}\right),$$

where $0 < \pi < 1$. However, the q quantile curves cannot be expressed as an explicit function and numerical methods need to be used to back out a value of u_2 from $C^{NM}\left(u_1 \middle| u_2\right) = q$ from each u_1 and *q*. If we set the N(0,1) distribution as the marginals for returns of stocks 1 and 2, then obtain and plot the *q* quantile curves for $q = 0.05, 0.25, 0.5, 0.75, 0.95$ when $\pi = 0.3,\ \rho_1 = -0.7,\ \rho_2 = 0.4$.

Exercise 4. Consider the slides "Option-based portfolios". Answer the following questions:

4.1) Check the graphs of the mean, volatility, skewness, and kurtosis on pages 16 to 19. Use a grid of strike prices ranging from 85 to 115, with a step size of 1. All parameter values can be found in the slides. The number of simulations is 100,000.

Hint: Run Matlab program **"cck_simuRet2.m**". This program uses "**BS.m**"

4.2) Repeat the previous analysis but now the correlation between the two asset returns is $\rho = 0.3$.

4.3) Repeat the same analysis as in section 4.2) but assuming a t copula with $\rho = 0.3$ and $v = 5$

4.4) Repeat the same analysis as in section 4.1) but assuming a bivariate Clayton copula with $\theta = 10$.

Exercise 5. Read up to and including section 3.1 of Berger's paper in *Economic Modelling* (58, 2016). Try to implement the simulation design in section 3.1 (specifically, see Table 2) by considering only as our data generating process (DGP) the bivariate Clayton copula under the three different parameter values and N(0,1) as marginal distributions. Estimate the bivariate distributions assuming different models: Gaussian, t and Clayton copulas with N(0,1) marginals. Evaluate the VaR forecast through backtesting criteria of unconditional coverage (Kupiec test).

Exercise 6. Referring to Lo (2007), verify equation (26) for the stop-loss policy. To do so, simulate the portfolio return process defined by equations (23), (24), and (25), and compare the simulated expected portfolio returns, for a selection of parameter values, with the corresponding entries reported in Table 1.

Exercise 7. (Optional) For each student, the oral exam will also include a very short question selected from one of the following two documents: *statSR.pdf* and *MarketTiming.pdf*. The format and scope of this question will be explained in class beforehand.